

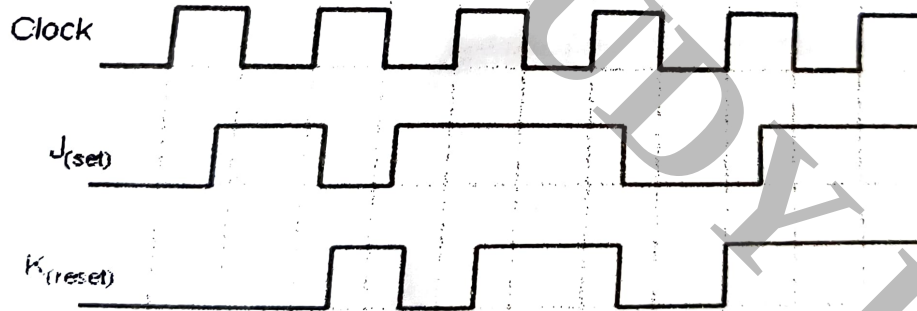
Class Test 2
CER3C4-Digital Electronics
Computer Engineering sec A and sec B

Time: 70 Min

Max Marks: 20

Note: Questions no. 1 and Q.2 are compulsory and solve any two from Q.3, Q.4 and Q.5.

- Q.1 simplify the following Boolean function, $f(W,X,Y,Z) = \sum m(2,6,8,9,10,11,14,15)$ using K-Map and Quine-McCluskey tabular method. 6**
- Q.2 Design positive edge trigger JK flip-flop circuit and explain its working and draw timing diagram for Q and Q' when JK and Clock signal is given below 6**



- Q.3 Design 4-bit magnitude comparator circuit using truth table and explain its working. 4**
- Q.4 What is the functioning of encoder and design Priority encoder and explain its working with the help of truth table. 4**
- Q.5 Define multiplexer and Design 8×1 multiplexer with the help of truth table and circuit diagram. 4**